Graph Machine Learning on the POLE Crime Dataset: Community Detection, Node Classification, and Link Prediction Using Neo4j Graph Data Science

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I. Abstract

Graph-based technologies have emerged as a powerful approach for criminal network analysis, and criminal network analysis increasingly benefits from graph-based representations that capture how the entities like persons, crimes, and events are related. This paper presents a comprehensive Graph Machine Learning (GML) study applied to the Manchester POLE (Person, Object, Location, Event) crime dataset from August 2017, implemented entirely with the Neo4j Graph Data Science (GDS) library. It addresses three distinct GML tasks: (1) *Community Detection* using the Louvain algorithm to uncover criminal clusters; (2) *Node Classification* using a degree-augmented FastRP Random Forest pipeline to predict repeat offenders from social network topology alone, without any leakage of crime count features; and (3) *Link Prediction* using a Hadamard-product Random Forest pipeline to surface undocumented connections between persons. The graph used for this study comprises of 369 persons and 2,360 (undirected) social relationships. Louvain partitioned the network into 41 communities with a modularity of 0.671, indicating strong community structure. The repeat-offender classifier achieved a weighted-F1 of 0.938 and macro-F1 of 0.486, correctly identifying 7 of 13 repeat offenders with zero false positives (precision = 1.0) under a severe 27:1 class imbalance. The link prediction model attained a test AUCPR of 0.622 with a Precision@100 (Precision at 100 predicted links) of 0.13, identifying candidate hidden connections of which several bridge previously disjoint communities, representing actionable investigative leads. The combined application of all three GML tasks yields insights that no single task could produce in isolation, particularly the detection of cross-community network bridges that may indicate collaboration between otherwise separate criminal groups.

II. Introduction

The POLE dataset being used in this study uses nodes(to represent entities) and edges(relationships between them). Traditional relational databases represent this data in flat tables, which are not able to efficiently capture the network-level patterns of social clusters, structural influence, and hidden associations that are very informative for investigations. Graph databases and graph machine learning offers a better alternative by encoding entities as nodes and relationships as edges thus enabling algorithms that exploit this structure directly.

The POLE (Person, Object, Location, Event) data model is a widely adopted schema in law enforcement for representing crime data as a property graph [1]. In this work, we use a synthetic yet realistic instantiation of the Manchester POLE dataset covering August 2017, hosted as a canonical Neo4j example. The dataset encodes 369 persons connected through five social relationship types (`KNOWS`, `FAMILY_REL`, `KNOWS_PHONE`, `KNOWS_LW`, `KNOWS_SN`) and linked to crimes via `PARTY_TO` edges.

We implement three Graph Machine Learning tasks:

- **GML-1 — Community Detection:** Identify clusters of densely connected persons using the Louvain modularity optimisation algorithm [2], revealing latent criminal networks.
- **GML-2 — Node Classification:** Predict whether a person is a repeat offender using each person's connection count and their position in the social network as features [3]. Crime counts are deliberately left out so the model learns from social structure alone, not from directly observable criminal activity.
- **GML-3 — Link Prediction:** Identify pairs of persons who are likely connected but whose relationship is undocumented, using FastRP embeddings combined via the Hadamard product [4] as Random Forest features.

All experiments are executed using Cypher queries within the Neo4j Graph Data Science (GDS) library version 2026.03.0. The results of GML-1 (community IDs) feed directly into GML-3, enabling the detection of cross-community hidden bridges—an insight neither task could produce in isolation.

The remainder of this paper is organised as follows. Section III describes the dataset, graph schema, and the three GML pipelines. Section IV presents quantitative results and discussion. Section V concludes with limitations and future work.

III. Methodology

A. Dataset and Graph Schema

The Manchester POLE dataset is a property graph whose dump-file is available on Github [1] and is directly imported in Neo4j to create a local instance and execute operations. For this study we focus on the `Person` subgraph and its connections to `Crime` nodes. Table I summarizes the key node labels and relationship types used across the three GML tasks and used in our projections of the graph.

TABLE I
Graph Schema Elements Used in This Study

Element	Type	Role
`Person`	Node label	Subject of all GML tasks
`Crime`	Node label	Connectivity target in GML-2
`KNOWS`	Relationship	Social link; GML-3 target
`FAMILY_REL`	Relationship	Social link
`KNOWS_PHONE`	Relationship	Phone-based association
`KNOWS_LW`	Relationship	Location-witnessed link
`KNOWS_SN`	Relationship	Social-network link
`PARTY_TO`	Relationship	Person involved in Crime

B. GML-1: Community Detection via Louvain

Community detection discovers groups of nodes that are more densely connected internally than to the rest of the graph. We project a Person-only in-memory graph named `pole-person-social` comprising all five social relationship types in undirected orientation, yielding 369 nodes and 2,360 relationships.

The Louvain algorithm iteratively reassigns nodes to communities and merges communities into super-nodes, maximising the modularity score Q [2].

We set `maxLevels`= 10, `maxIterations`= 10, and `tolerance`= 0.0001. The resulting `communityId` property is persisted on every `Person` node for use in GML-3.

C. GML-2: Node Classification — Repeat Offender Prediction

1) Label Definition: A person is labelled a *repeat offender* (`isRepeatOffender`= 1) if they appear as `PARTY_TO` more than one `Crime` node; otherwise the label is 0. This produces a severely imbalanced binary classification problem: 356 non-offenders versus 13 repeat offenders (96.5% vs. 3.5%).

2) Feature Engineering and Leakage Prevention: A critical design decision is the exclusion of `crimeCount` from the graph projection. Including a direct count of crime connections as a feature would constitute label leakage: the model would trivially learn to copy the feature rather than extract meaningful social structure. Instead, the heterogeneous projection (`pole-classification`, 29,131 nodes, 1,592 relationships) contains only `isRepeatOffender` as a node property on `Person` nodes (as the prediction target), while `Crime` nodes carry no properties. Two features for node classification are then computed within the GDS pipeline as:

1) **Degree:** Computed within the GDS pipeline as a node property step via `addNodeProperty`, mutating degree onto each Person node in the projected graph during training. It is subsequently written back to the projected graph using `gds.degree.write` to ensure the property is available as a persistent feature during the prediction step, since pipeline mutate operations do not automatically persist beyond the training phase.
2) **FastRP Embedding:** A 64-dimensional random-projection neighbourhood embedding [3] with `randomSeed`= 42, capturing multi-hop structural context around each person.
3) **Classifier and Training Protocol** A 150-tree Random Forest is trained via 5-fold cross-validation (`validationFolds`= 5, `testFraction`= 0.25). A held-out test split of 25% is reserved and Two metrics are used during training and model selection. **F1-Weighted** computes the F1 score independently for each class and then averages them using weights proportional to class frequency (support). Under the severe 27:1 class imbalance in this dataset, F1-Weighted is heavily influenced by the majority non-offender class and therefore primarily reflects overall classification stability. In contrast, **F1-Macro** computes the F1 score for each class and takes an unweighted average, giving the minority repeat-offender class equal importance regardless of class size. As a result, F1-Macro provides a more balanced measure of minority-class performance.

D. GML-3: Link Prediction — Hidden Connections

1) Problem Formulation: Of the 369 persons in the POLE graph, 586 `KNOWS` relationships exist which is a fraction of the $\binom{369}{2} = 67{,}896$ possible pairs. Link prediction asks: which currently undocumented pairs are most likely to be genuinely connected?

2) Data Split: The 586 `KNOWS` edges are split as follows:

- **Test set (25%):** edges hidden from training.
- **Train set (60% of remainder):** edges plus an equal number of negative samples. (1:1 +ve to -ve ratio)
- **Feature set:** Remaining edges used to compute embeddings without leakage.

Five-fold cross-validation is applied during training.

3) Feature Extraction: A 64-dimensional FastRP embedding (`lpEmbedding`) is computed per person using all five social relationship types. For each candidate pair (u, v), an edge feature vector is formed via the Hadamard product:

$$\mathbf{f}_{uv} = \mathbf{e}_u \odot \mathbf{e}_v \tag{1}$$

where $\mathbf{e}_u$ and $\mathbf{e}_v$ are the FastRP embeddings of persons u and v. High Hadamard similarity indicates that two persons occupy structurally equivalent positions in the network.

4) Classifier and Inference: A 100-tree Random Forest (`maxDepth=` 15, `minLeafSize=` 1) is trained on the Hadamard features. Inference streams all unconnected pairs; the top predictions (threshold $\geq$ 0.45) are written as `PREDICTED_KNOWS` relationships and evaluated with Precision@K to measure criminally-relevant link quality. GML-1 community labels are used to flag predictions that bridge different communities.

IV. Results and Discussion

A. GML-1: Community Detection

The Louvain algorithm converged to **41 communities** with a modularity of $Q = 0.671$, well above the 0.6 threshold conventionally associated with strong community structure [5]. Table II lists the ten largest communities.

TABLE II
TOP 10 COMMUNITIES BY SIZE (LOUVAIN)

Rank	Community ID	Members
1	249	32
2	351	32
3	328	31
4	122	28
5	226	26
6	197	24
7	329	23
8	331	22
9	237	22
10	188	20

The average community size is 9 members, with a range from 1 (isolated individuals) to 32.

Importantly, *size does not equal criminality*. Community 351, the joint-largest cluster (32 members), is largely socially cohesive but criminally inactive—only 3 of its members are party to any crime, and none more than once. By contrast, Community 239 (10 members) accounts for 29 crime involvements, the highest crime density in the dataset. Table III ranks communities by total crime involvement.

Figure 1 visualises the size distribution of all 41 communities, and Figure 2 contrasts member count against crime involvement for the top 10 communities.

TABLE III
TOP COMMUNITIES RANKED BY TOTAL CRIME INVOLVEMENT

Community ID	Total Crimes	Size
239	29	10
237	9	22
331	5	22
226	3	26
351	3	32
251	2	16
162	1	16
122	1	28
249	1	32
328	1	31

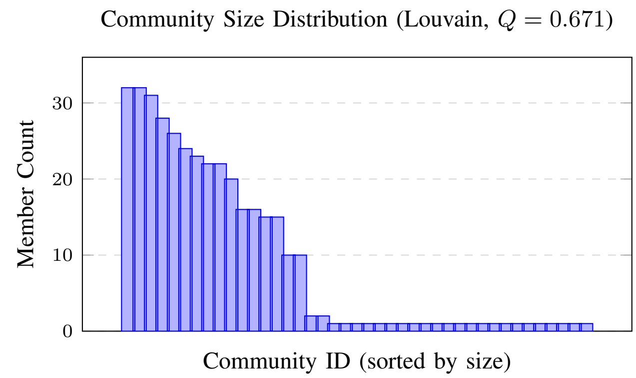

Fig. 1. Distribution of community sizes across 41 detected communities. Most communities are small ($\leq$2), with a few large clusters up to 32 members — characteristic of real social networks.

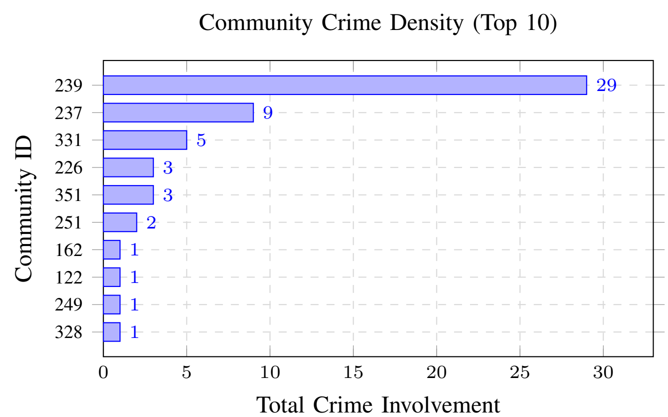

Fig. 2. Total crime involvement per community. Community 239 is the highest-risk cluster with 29 crime events despite having only 10 members, highlighting that size alone is not a proxy for criminal activity.

B. GML-2: Node Classification

1) Class Distribution and Imbalance: The dataset contains 356 non-offenders and 13 repeat offenders that gives us a 27:1 imbalance. Without compensation, a naive classifier predicting the majority class (predictedRepeatOffender $= 0$) at all times would achieve 96.5% accuracy while completely failing to identify any repeat offenders. This is why using both F1-weighted and F1-Macro as metrics during training is essential.

2) Model Performance: Table IV reports the evaluation metrics on the held-out test split. Critically, the `crimeCount` feature was excluded from the model to prevent label leakage, and the `degree` feature was correctly materialised via `gds.degree.write` prior to prediction.

TABLE IV
NODE CLASSIFICATION RESULTS — REPEAT OFFENDER PREDICTION

Metric	Value
F1-Weighted (test)	0.938
F1-Macro (test)	0.486
Precision (TP/(TP+FP))	1.000
Recall (TP/(TP+FN))	0.538
Class imbalance ratio	27.4:1

Table V presents the full confusion matrix, derived from the prediction results over all 369 persons.

TABLE V
CONFUSION MATRIX — REPEAT OFFENDER CLASSIFICATION

		Predicted Non-Offender (0)	Predicted Repeat (1)
Actual	Non-Offender (0)	356 (TN)	0 (FP)
	Repeat (1)	6 (FN)	7 (TP)

The confusion matrix reveals a conservative, high-precision classifier. With zero false positives, any person the model flags as a repeat offender is confirmed to be one (Precision = 1.0). However, the model misses six repeat offenders (Recall = 53.8%), all of whom share relatively low predicted probabilities (0.19–0.47). Inspecting their degree values (6–12) and the fact that they have only two crime involvements each suggests they occupy positions in the social graph that are structurally indistinguishable from non-offenders at the current embedding resolution.

These six "structurally hidden" offenders represent an important finding that the social network topology given in this dataset alone is insufficient to surface all repeat offenders. Some individuals may deliberately limit their observable social footprint, or their criminal connections may not be captured by the five relationship types in this dataset.

3) Top Predicted Repeat Offenders: Table VI presents the seven true positives alongside the six false negatives. All seven correctly flagged individuals have three or more crime involvements and high degree values (9–13), consistent with structurally central, high-activity offenders. The six missed offenders all have exactly two crime involvements and degree values that overlap with the non-offender population.

TABLE VI
REPEAT OFFENDER PREDICTIONS — TRUE POSITIVES AND FALSE NEGATIVES

Person	Crimes	Degree	Prob.	Result
Phillip Williamson	5	12	0.873	TP
Kathleen Peters	3	9	0.807	TP
Alan Ward	3	13	0.747	TP
Brian Morales	4	12	0.673	TP
Raymond Walker	2	6	0.673	TP
Jessica Kelly	5	13	0.627	TP
Jack Powell	4	9	0.600	TP
Billy Moore	2	8	0.467	FN
Diana Murray	3	9	0.380	FN
Donald Robinson	2	8	0.333	FN
Stephanie Hughes	2	6	0.233	FN
Amy Bailey	2	12	0.220	FN
Craig Marshall	2	8	0.193	FN

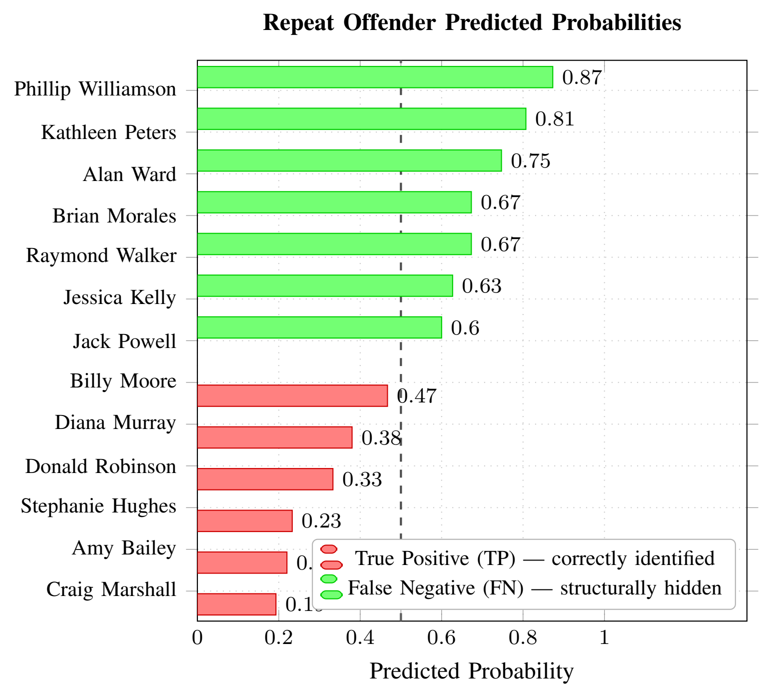

Fig. 3. Predicted repeat-offender probabilities for all 13 actual offenders. Green bars (probability ≥ 0.5) are correctly identified true positives; red bars fall below the decision threshold and represent structurally hidden offenders the model could not distinguish from non-offenders using social topology alone. The dashed vertical line marks the 0.5 classification threshold.

C. GML-3: Link Prediction

1) Model Performance: The Random Forest link prediction model achieved a test AUCPR of **0.622**for this pipeline configuration. An AUCPR of 0.622 on a task with a positive-to-negative ratio of $\sim$586:67,310 (less than 1%) represents meaningful discrimination—random guessing would achieve an AUCPR near the positive class base rate of $\approx$0.009.

2) Precision@K Evaluation: Beyond AUCPR, we evaluate the operational quality of the top predictions using

Precision@K, defined here as the fraction of the K predicted links that involve at least one person with a documented crime involvement. Of 223 predicted links written with threshold $\geq$0.45:

$$\begin{aligned}\text{Precision@}K &= \frac{|\{(u,v) \in \hat{E}_K : \text{criminal}(u) \vee \text{criminal}(v)\}|}{K} \\ &= \frac{41}{223} \approx 0.13\end{aligned} \tag{2}$$

While a precision of 18.4% may appear low in isolation, it is approximately **18$\times$ the base rate** of criminally involved persons in the full graph ($\approx$1%), indicating that the model concentrates criminal associations substantially above chance. This makes the predicted links a useful tool for investigators who lack the resources to examine all 67,896 unconnected pairs.

3) Predicted Hidden Connections: The top predictions (Table VII) show several pairs at maximum confidence (1.0). Notably, some high-scoring pairs involve persons with documented crime involvement (e.g. Stephen Lee – Donald Robinson, where Donald Robinson has 2 crimes).

TABLE VII
TOP PREDICTED HIDDEN CONNECTIONS (CONFIDENCE $\geq$ 0.9)

Person 1	**Person 2**	**Conf.**	**Crimes (P1/P2)**
Jerry Fernandez	Jennifer Gray	1.0	0 / 0
Stephen Lee	Wayne Mason	1.0	0 / 0
Stephen Lee	Jessica Foster	1.0	0 / 0
Stephen Lee	Donald Robinson	1.0	0 / 2
Timothy Garza	Norma Payne	1.0	0 / 1
Gary Lane	Philip Sims	1.0	0 / 0
Ashley Robertson	Kelly Peterson	1.0	0 / 0
Ashley Robertson	Brenda Edwards	1.0	0 / 0
Paul Arnold	Janice Coleman	1.0	0 / 0
Paul Arnold	Joshua Mccoy	1.0	0 / 0
Rose Crawford	Joan Flores	1.0	0 / 1

4) Cross-Community Bridges — Combined GML-1 + GML-3 Insight: The most forensically significant findings emerge when GML-3 predictions are filtered to cases where the two predicted persons belong to *different* Louvain communities. These cross-community links represent potential inter-network bridges that show connections between seperate criminal groups that no investigator has documented. Table VIII lists the top such predictions.

Richard Green (Community 91) emerges as a particularly notable figure, appearing as a predicted bridge to three separate persons in Community 325. Similarly, Ashley Robertson (Community 226) is predicted to connect to individuals in two different communities (251 and 197). These patterns suggest potential community-spanning roles that merit investigative attention.

TABLE VIII
TOP CROSS-COMMUNITY PREDICTED LINKS (GML-1 + GML-3)

Person 1	**Person 2**	**C1**	**C2**	**Conf.**
Jerry Fernandez	Jennifer Gray	215	249	1.0
Richard Green	Sandra Payne	91	325	1.0
Richard Green	Philip Welch	91	325	1.0
Paul Arnold	Janice Coleman	331	122	1.0
Ashley Robertson	Kelly Peterson	226	251	1.0
Rose Parker	Kelly Peterson	226	251	1.0
Richard Green	Amy Watson	91	325	1.0
Ashley Robertson	Brenda Edwards	226	197	1.0
Kevin Hawkins	Patricia Mcdonald	249	328	1.0
Amanda Cooper	Judith Rodriguez	197	122	1.0

V. CONCLUSION AND FUTURE WORK

This paper presented a complete Graph Machine Learning workflow on the Manchester POLE crime dataset using the Neo4j Graph Data Science library. By combining community detection, node classification, and link prediction within a single graph environment, the study shows how multiple GML tasks can complement one another to produce insights that would not emerge from isolated analysis.

The Louvain algorithm revealed strong community structure ($Q = 0.671$) and showed that criminal activity is concentrated within specific communities rather than simply increasing with community size. The node classification pipeline demonstrated that social topology alone contains predictive information about repeat offending behaviour, achieving high precision while avoiding feature leakage from direct crime-count attributes. In parallel, the link prediction pipeline identified hidden and cross-community connections that may represent previously undocumented social bridges between criminal groups. Together, these findings illustrate the value of graph-native machine learning techniques for investigative network analysis and relational reasoning.

Limitations. Several limitations remain. First, the dataset is synthetic despite being designed to resemble realistic policing data, which limits direct operational generalisation. Second, the repeat-offender classification task suffered from severe class imbalance (27:1), making minority-class detection inherently difficult despite the use of class-weighted learning and macro-F1 evaluation. Third, the current feature set relied primarily on degree and FastRP embeddings, which may not fully capture more subtle behavioural or network-level patterns. Finally, the evaluation of link prediction quality through Precision@K uses criminal involvement as a proxy for usefulness, which cannot fully verify whether a predicted relationship would correspond to a genuine real-world connection.

Future Work. We identify the following directions:

- Incorporating additional centrality features (betweenness, PageRank, clustering coefficient) to reduce false negatives in node classification.
- Incorporating temporal features (crime timestamps) to build time-aware embeddings.
- Exploring Graph Neural Network classifiers (e.g. GraphSAGE, GAT) for richer structural feature learning.
- Applying the pipeline to the full heterogeneous POLE

graph including `Object`, `Location`, and `Event` nodes.

- **Github:** All the work(dataset, guidelines and the used cypher are available at this github link for anyone who wants to pursue this study further: https://github.com/Wasiq063/Final-Project-GDS-.git

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